

Machine Learning and Deep Learning Methods for Intrusion Detection Systems in IoMT: A survey

Yahya RBAH

*Artificial Intelligence & Data Science
& Emerging Systems Laboratory
Sidi Mohamed Ben Abdellah University
of FES
rbah.yahya@gmail.com*

Mohammed FATTAH

*Image Laboratory
University of Moulay Ismail
of Meknes
fattahm@gmail.com*

Benaissa BERNOUSSI

*Artificial Intelligence & Data Science
& Emerging Systems Laboratory
Sidi Mohamed Ben Abdellah University
of FES
bbernou@gmail.com*

Mohammed MAHFOUDI

*Artificial Intelligence & Data Science
& Emerging Systems Laboratory
Sidi Mohamed Ben Abdellah University
of FES
mahfoudi.mohammed@gmail.com*

Said MAZER

*Artificial Intelligence & Data Science
& Emerging Systems Laboratory
Sidi Mohamed Ben Abdellah University
of FES
mazersaid@gmail.com*

Younes BALBOUL

*Artificial Intelligence & Data Science
& Emerging Systems Laboratory
Sidi Mohamed Ben Abdellah University
of FES
balboulyounes@gmail.com*

Moulhime ELBEKKALI

*Artificial Intelligence & Data Science
& Emerging Systems Laboratory
Sidi Mohamed Ben Abdellah University
of FES
Moulhime.elbekkali@usmba.ac.ma*

Abstract— The integration of healthcare-related sensors and devices into IoT has resulted in the evolution of the IoMT (Internet of Medical Things). IoMT that can be viewed as an improvement and investment in order to meet patients' needs more efficiently and effectively. It is progressively replacing traditional healthcare systems, particularly after the worldwide impact of COVID. IoMT devices have enabled real time monitoring in the healthcare field, allowing physicians to provide superior care while also keeping patients safe. As IoMT applications have evolved, the variety and volume of security threats and attacks including routing attacks and DoS (Denial of Service), for these systems have increased, necessitating specific efforts to study intrusion detection systems (IDSs) for IoMT systems. However, IDSs are generally too resource intensive to be managed by small IoMT devices, due to their limited processing resources and energy. In this regard, machine learning and deep learning approaches are the most suitable detection and control techniques for IoMT device-generated attacks. The purpose of this research is to present various methods for detecting attacks in the IoMT system. Furthermore, we review, compare, and analyze different machine learning (ML) and deep learning (DL) based mechanisms proposed to prevent and detect IoMT network attacks, emphasizing the proposed methods, performances, and limitations. Based on a comprehensive analysis of current defensive security measures, this work identifies potential open research related challenges and orientations for the actual design of those systems for IoMT networks, that may guide further research in this field.

Keywords— *IoMT, Security, Privacy, intrusion detection system (IDS), Machine Learning (ML), Deep Learning (DL).*

I. INTRODUCTION

The advancement of technologies including 5G, artificial intelligence (AI), and IoT has resulted in the evolution of emerging paradigms including smart grids, smart cities, smart factories, and smart healthcare, all of which have contributed to improving global life quality in a cost-effective and

efficient manner [1]. The incorporation of medical devices into the IoT has contributed in the emerging of the IoMT (Internet of Medical Things). As the new era of digitalized healthcare, known as Healthcare v4.0, IoT enabled devices have been deployed in multiple medical fields, particularly with the widespread use of wireless medical sensors, devices, and robots [2].

The IoMT has improved a variety of medical applications, including remote patient monitoring, early diagnosis, medical emergencies, and real-time monitoring. It is reducing both unnecessary visits to the hospital and the burden on the healthcare system by connecting physicians to patients and enabling the transfer of medical information over a secure network. In addition, it can help detect any medical problems at an early stage and thus preserve the life and health of patients [3]. Through the use of technology, IoMT has a great potential to provide accuracy in diagnosis, reducing errors and lowering the cost of care by enabling patients to transmit healthcare data to physician. Nowadays, this is basically required because of the effect of the worldwide pandemic, COVID-19 had also made it obligatory to make the medical services accessible online to any remote place, thus decreasing in-person medical consultations that prevents the propagation. In fact, the medical actuators and sensors are employed as implantable or small wearable devices within the body area network. Rather than keeping the patients in the hospitals, such devices are able to continuously monitor the patient's health in real time, while providing better physical flexibility and mobility. In addition, medical robots can also be used as surgical robots and hospital robots, as they are capable of performing small surgical operations accurately [2].

However, IoMT is subject to different types of attacks, whether they are internal or external, passive or active. During the transmission of critical and sensitive data from medical devices to the fog and then to the cloud, an attacker can interfere with this transmission because of weak

authentication mechanisms and poor design. As a result, an intruder can carry out a variety of cyber-attacks, including jamming, spoofing, ransomware, and DoS attacks [4]. Moreover, any threat to the security of these systems can lead to a serious problem, for instance a wrong diagnosis or delayed interactions. This results in a breach patients' privacy, health problems, and in extreme cases, death. For example, when an interconnected insulin pump gets an invalid command, it can inject an over-dose of insulin into patient's blood. Similarly, cardioverter defibrillators may automatically deliver electrical shocks when they detect that the heart is not beating properly [5].

The resource limitations, such as compute and storage, of IoMT devices make traditional systems unsuitable in particular. In fact, systems that are designed for IoMT environments must operate within strict limits, including low processing capacity, fast response, and processing of large volumes of data [6]. To avoid such situations, machine learning (ML) and deep learning (DL) methods have been employed to develop an intrusion detection system (IDS) for IoMT networks. An IDS can monitor network traffic to protect users from malicious actions that compromise the confidentiality, integrity, and availability of the given information system. [7].

In light of the aforementioned issues, this study focuses on investigating IDS for IoMT systems, both by conducting a brief overview of different variants of IDS to provide insight into their specifications and capabilities, regarding resources utilization including CPU, power, and bandwidth usage, in addition to the trade-off with metrics including detection accuracy, false positives, and false negatives. Additionally, we provide a comparison of the main ML and DL based approaches, proposed for IDSs in IoMT networks. Specifically, we emphasize open research challenges that may lead to future research avenues, while also attempting to identify and discuss the limitations of current solutions and highlighting future directions.

II. RELATED WORKS

Intrusion detection has been a peculiar issue confronting researchers in the security domain. Machine learning techniques, especially deep learning, have proven to be among the most suitable techniques for combating network intrusions. In recent years, various surveys have conducted on intrusion detection methods based on these approaches for protection against IoMT networks or compromised IoMT devices [8] [9] [10] with little regard for their IoMT applications. In [2], the authors presented a detailed study on cyber-attacks target the domains of IoMT and healthcare, as well as they discuss the security measures that are adopted in such domains. A study in [11] examine the present state of privacy and security of the Medical IoT, in terms of the medical IoT architecture, the authors classify privacy and security attacks and propose solutions and countermeasures to avoid these attacks. In a recent Systematic Literature Review (SLR) published in [12], authors discussed the privacy and security concerns of IoMT and their countermeasures by ML methods. Moreover, they try to highlight the current methods' limitations and aim to identify potential solutions to them. Another study in [13] Provide an analysis of ML and DL-based IDS suitable for detecting IoMT systems related to cyber-attacks by involving the design choices, limitations,

advantages, and detection approaches. However, most existing works related to IoT security has not primarily concentrated on the implementation of ML or DL methods as detection to secure the IoMT environment. Therefore, a detailed study of ML as well as DL based methods for IDS in IoMT networks still requires a classification and analysis of current work as well as an investigation and discussion of open questions in this field.

III. INTRUSION DETECTION SYSTEMS (IDS)

The process of the monitoring, detection and identification of malicious activity on network or computer system is called intrusion detection [14]. A generic IDS consists of three components, which are the monitoring, the detection and the response. The monitoring component analyzes the traffic flow behavior. The detecting component detects any suspect behaviors and notifies the reacting element of any occurrences detected. The response component triggers an alert or informs the network manager. Depending on its working principles and analysis methods, An IDS can be based on signatures, anomalies or specifications. while it can be categorized as network-based, host-based, or hybrid IDS depending on the target location, as shown in Fig. 1 [2].

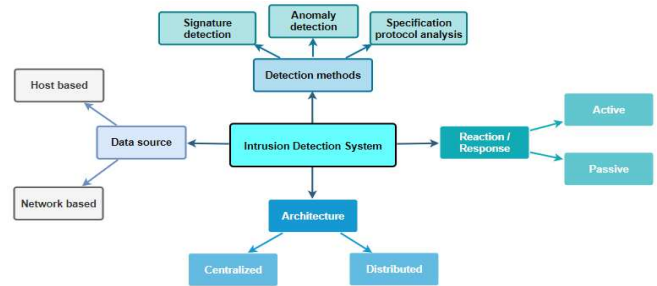


Fig. 1. A classification of different IDS categories.

Signature-based systems, also known as misuse detection methods, detect previously known attacks by comparing their signatures to the data being analyzed [15]. They are effective in contexts where most attack patterns are already known. Their main limitation is the need for regular signature database updates, as well as the inability to detect zero-day attacks and unknown threats [16]. In contrast, anomaly-based IDSs use sophisticated decision techniques like ML to detect threats based on the characteristics of the traffic flow. They monitor the usage behavior deviation from the normal functionality [17]. One advantage of using anomaly-based schemes is the ability to be effective in detecting both unknown and zero-day attacks, but they can generate high false-positive outcomes. [18]. On the contrary, specification-based (hybrid systems), for identifying attacks, are using a combination of the two variants mentioned earlier in this section, it employs a halfway technique between signature-based IDS, in which pre-defined rules are used to identify known attacks, as well as anomaly-based IDS, in which the normal behavior can be set to discover any unusual activity, even unknown, unlike signature-based IDS [19]. In specification-based IDS, the developers manually set the constraints and characteristics that depict normal operation behavior, in contrast to anomaly-based IDS, which are generated automatically. Therefore, the algorithm of intrusion detection can identify suspicious activity by discovering the deviation between the two patterns [20]. Approaches detection based on specification and signature need low overhead in comparison to the anomaly-based

methods. However, due to computational power limitations and the large numbers of interconnected devices, a conventional IDS based on anomalies is not effective in the IoMT [2].

In addition, IDS are categorized into two classes based on their deployment, namely network-based IDSs and host-based IDSs [21]. In the host-based IDSs, the detector system is deployed and installed within a particular host for monitoring the behaviors of a specific system, including analysis of inbound and outbound traffic, system calls, file system access details, and so on [16]. These systems can only monitor the network passing through the host on which they are implemented. The detection of botnets and malware are the two main types of host-based IDS [22]. Threats in other parts of the system are undetectable. Network-based IDSs are strategically placed inside a network, such as at the gateway node or in a decentralized way, to identify any threat on the network's hosts. Because network-based IDS are monitoring all traffic passing over the network, they must analyze traffic quickly and drop as few traffic data packet as possible [16].

IDSs are also classified into two categories: active and passive. Active IDS also known as IPS (Intrusion Prevention System) [23]. IPS is designed to monitor activities at the system or network level to identify attacks and deliver corrective actions in real time without requiring an administrator's intervention. On the other side, passive IDSs can only detect attacks. It alerts the administrator if any attack is detected and cannot perform any corrective action by itself. By default, the majority of IDSs are passive [16].

Most intrusion detection systems have a basic common design that contains: (1) a data collecting module that gathers data which may contain proof of an intrusion, (2) an analyze module that identifies threats after handling this data, and (3) a process to report an intrusion [13]. As shown in Fig. 2. In data collection subsystem, input data from each component of the IoMT systems can be collected and analyzed to find the behavior pattern of the interaction, and thus detect the malicious behaviors at an early stage. The analytics module could be deployed with various methods and techniques, although, approaches based on ML and DL are more appropriate and dominating for the data examination in order to learn the benign and abnormal behaviors resulting from the interaction of IoMT systems and devices in IoMT environments. In addition, ML/DL techniques are able to predict novel attacks, that are often significantly different than previous attacks, as ML/DL methods are capable of intelligently predicting the future unknown threats by learning from the available legitimate samples [24].

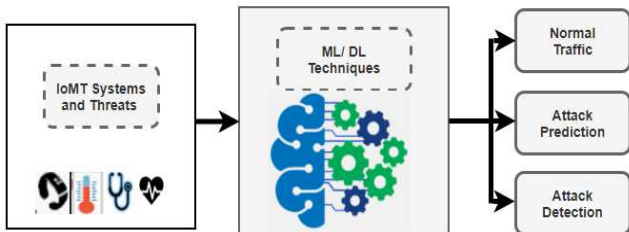


Fig. 2. Role of ML/DL based IDS for IoMT system.

IV. METRICS AND DATASETS USED FOR EVALUATION

Once the IDS model is built using the above methods, it must be assessed using specialized IDS tools such as Snort or general-purpose languages and programming environments

like Python and MATLAB. In addition, IDS models can be evaluated in the real environments of the hosts or networks being studied [25].

A. Evaluation metrics of IDS for IoT

It is critical to understand the standardized measures used in the IDS development and research community in order to evaluate a proposed IDS as well as compare its performance with current IDS methods. Various metrics (or parameters) that were used for evaluating an IDS on the IoT. They can be classified into two categories: metrics related to detection and those related to resource utilization as following:

1) Detection Related Metrics

The classifier must be trained utilizing IDS dataset training data and tested using IDS dataset testing data [26]. An IDS result can be categorized as follows:

- TP (True Positive) is the amount of malicious data in network traffic on the IoMT, which are labeled as suspicious.
- TN (True Negative) represents the amount of benign data in the IoMT network traffic, which are expected to be benign.
- FP (False Positive) refers to the benign or normal samples in the IoMT network traffic, which are misclassified as malicious samples or attacks.
- FN (False Negative) correspond to malicious samples in the IoMT network traffic, which are misclassified as normal.

The Confusion Matrix can be used to define the overall output for a given classification model. It provides a global vision of the classification performances, composed of four main metrics: false negative, false positive, true negative, and true positive [27]. As illustrated in Fig. 3.

		Actual	
		Positive	Negative
Predicted	Positive	True Positive	False Positive
	Negative	False Negative	True Negative

Fig. 3. The confusion matrix.

Confusion Matrix that can be represented in a binary or multi-class manner. It is useful for calculating accuracy, precision, recall as well as the AUC-ROC curve values [26].

Other significant metrics for IDS performance measures are sensitivity, recall, precision, F1 score, and accuracy. Table I provides the evaluation parameters calculated using the confusion matrix [28] [29].

Finally, another essential factor for evaluating classification performance is the ROC (Receiver Operating Characteristics) curve and AuC (Area under the ROC). The ROC curve represents a graph of the TPR (true positive rate) and FPR (false positive rate) that is defined as the area under the ROC curve (AuC) and has the range from 0.5 to 1.0. An AuC value equal to 1 means that an IDS scheme performs perfectly in terms of classification and an AuC value lower than 0.5 indicates that the detection performance is very poor [27].

2) Resource Usage Metrics

Resource utilization measures are also called performance metrics, since they evaluate the performances of the system where the IDS is located. These metrics are used to examine the impact of the IDS on the overall system. In general, the resource usage metrics are categorized as follows: memory utilization, CPU utilization, and energy overhead. The assessment of an IDS in an IoMT network using these metrics is very important [14]. The following measures are considered in this context:

- CPU Usage: refers to the percentage of CPU computation used by the IDS.
- Memory Usage: represent the proportion of available memory used by the IDS. In this instance, the memory is usually the ROM or RAM used by the device carrying intrusion detection.
- Energy Overhead (power overhead): is an important precondition for the IDS to be deployed in the IoMT. Both metrics of CPU and memory usage directly impact on the power overhead of a given IoMT node.

TABLE I. THE EVALUATION METRICS CALCULATED USING THE CONFUSION MATRIX

Performance Metrics	Description	Formula
<i>Accuracy</i>	Number of corrects predictions on total predictions.	$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$
<i>Precision</i>	A metric for calculating the accuracy of ML model in identifying the total number of true positives among all predicted positives.	$Precision = \frac{TP}{TP + FP}$
<i>Recall/Sensitivity</i>	A metric for calculating the accuracy of ML model in identifying the total number of positives among all true positives.	$Recall = \frac{TP}{TP + FN}$
<i>Specificity</i>	The proportion of true negative events that were predicted to be negative.	$Specificity = \frac{TN}{FP + TN}$
<i>F-1 Score (Measure)</i>	A harmonic average of recall and precision metrics to accurately evaluate performance of the model.	$F1 - score = 2 * \frac{Precision * Recall}{Precision + Recall}$
<i>False Positive Rate (FPR)</i>	The number of misclassified negative samples in relation to the total of negative instances.	$FPR = \frac{FP}{FP + TN}$
<i>False Negative Rate (FNR)</i>	It is the proportion of misclassified positive samples to total positive samples.	$FNR = \frac{FN}{TP + FN}$

B. Datasets used for evaluation

The existence of label datasets is a factor that directly influences the efficiency of IDS methods. Although those datasets are expensive, they improve the efficiency of the IDS methods [30]. In order to achieve this goal, research institutions make some datasets accessible that contain a number of "fingerprint" anomalies captured on the field. Several standard data sets have been provided and used in cyber-security attack detection in this context, such as CICIDS2017 [31], DARPA [32], UNSW-NB15 [33], NSL-KDD [34], and KDD-99 [35]. These datasets carry normal traffic with various sorts on security related attacks such as DoS, U2R (User-to-Root), R2L (Remote-to-Local), PROBE attacks etc. However, the attack data that are generated by IoMT devices are diverse, and the trained models using the non-IoMT dataset might initially provide high accuracy on the trained dataset. Therefore, it is critical to train the ML/DL based models of these systems on a specific dataset for the IoMT environment. As a result, several researches have been conducted to generate datasets used for the evaluation of IDS in IoT security-related proposals. Such as ISOT [36], ToN_IoT [37], The Bot-IoT dataset [38], IoT-23 [39], and IoT Routing Attack Dataset (IRAD) [40].

V. MACHINE LEARNING TECHNIQUES FOR IDS IN IOMT

Over the last decades, Machine Learning algorithms have been widely employed in several application domains like speech recognition, bioinformatics, spam detection, intrusion detection, etc. [41]. ML methods are used for constructing the

computational model in a cost-effective way. In addition, they improve efficiency and reliability. ML constructs the models by processing the data automatically, accurately and rapidly. It enables access to the massive amounts of data collected from sensors and the extraction of information required from the data collected [42]. Given the significance and complexity of an IDS, a variety of machine learning techniques and algorithms are used to offer practical IDS methods with high rate of true negatives and low false positive rates [25].

Machine learning techniques are commonly categorized into two classes, Known as Unsupervised Learning and Supervised Learning. Various machine algorithms have been developed in recent years. Popularly used algorithms include SVM (Support Vector Machine), RF (Random Forest), DT (Decision Tree), ANN (Artificial Neural Network) and EL (Ensemble Learning) [4] [17] [43]. Many researchers are interested in working on IoMT security issues using ML models due to recent advances made in ML technology. Table II provides a brief summary of the research work done to propose IDS for IoMT using numerous ML-based methods.

VI. DEEP LEARNING TECHNIQUES FOR IDS IN IOMT

With the drastic growth in the data complexity and volume of network traffic, IDS based on conventional ML with a shallow architecture is inappropriate for the IoT context involving billions of IoT devices [27]. Deep Learning appears to be one of the most appropriate and powerful methods to be

implemented to various IoT security measures in this context [44]. In addition, DL is able to automatically model complicated features sets from the samples data. A further advantage of Deep Learning based approaches is their

capacity to enable deep binding in IoT based networks [45]. This allows for automatic interventions among IoT-based systems without human intervention to execute collaborative assigned functions.

TABLE II. CURRENT ML-BASED IDS FOR IOMT.

Ref/year	Description	Algorithms	Attacks addressed	Dataset	Accuracy rate	Limitations
[46] 2017	A distributed IDS to secure WBANs using autonomous mobile-agents.	SVM, KNN, NBC, DT and RF.	Data Falsification, DoS, Passive Listening.	Simulated data.	97.21%.	-High FPR. -High training duration for certain techniques.
[47] 2017	A technique for identifying and detecting attacks on implanted medical devices (IMD) using ML methods.	SVM, K-means and decision tree.	Resource Depletion Attacks.	Simulation dataset.	Varies from 87.7%, to 90.37% for different algorithms.	-There is no consideration for usage memory or battery life. -Fails to detect insider threats. -There is no validation.
[48] 2019	A federated learning-based IDS for attacks detection in Medical Cyber Physical Systems (MCPS).	Federated Learning.	DoS, data injection and data modification.	MIMIC dataset.	99.0%.	-In certain cases, accuracy has been reduced while FPR has been increased. -Failure to repel adversarial attacks.
[49] 2019	A real-time model based on ML methods for detection and classification of ransomware attacks in integrated clinical environments (ICE).	OCSVM (One-Class SVM), Naïve Bayes, and NFV/SDN.	Ransomware detection: Petya, WannaCry and BadRabbit.	Self-created dataset.	92.32% for OC-SVM and 99.99% for Naive Bayes.	-No calculation of performance overhead. -Not lightweight.
[50] 2019	An IDS based on Stacked Autoencoder for detecting malicious activities in healthcare systems.	k-NN, NB, SVM, and XGBoost.	Counterfeit attack, DoS attack, replay attack, and tamper attack.	Self-collected data.	97.83% for XGBoost.	-No light weight. -High overhead training.
[51] 2019	HealthGuard, a ML data-driven security model for detecting threats in smart healthcare systems.	k-NN, RF, DT, and ANN.	False data injection, tampered medical devices, and DoS attack.	Physiological data collected from IoMT devices.	Varies from 90-93% for different algorithms.	-Not real-time. -CPU usage and memory are not taken into account.
[52] 2019	FBAD a fog-based IDS using the EOS-ELM (Ensemble of Online-Sequential Extreme Learning Machine) method to identify malicious activities.	OS-ELM algorithm.	U2R, R2L, Probe and DoS.	NSLKDD dataset.	98.19 %.	-CPU usage and memory are not taken into account. -Not real-time.
[53] 2019	A secure IDS for a secure remote health system that protects wireless sensor networks from unauthorized access as well as network attacks.	SVM.	DoS and U2R.	NSL-KDD dataset.	97.2%.	-Extreme memory overhead. -Outdated dataset. -In misuse analysis, there is a poor detection rate.
[5] 2020	A security system that detects anomalies by using network flow and patient biometrics metrics.	RF, SVM, ANN, and KNN.	MITM attack.	Real healthcare dataset.	Varies from 88.75% to 92.44% for different techniques.	FPR, CPU utilization, and memory are not taken into account.
[3] 2020	An IDS based on various mobile agents to detect intrusions using ML methods for IoMT.	SVM, KNN, DT, RF, and NB.	DoS attacks, Data Falsification and Fabrication, Privacy/Data breach.	Self-collected data.	99.8% for network, 97.93% for host intrusion detection.	-Memory, battery, and CPU utilization are not taken into account.
[54] 2021	A neural network-based system for detecting and disabling suspect devices in the IoT-enabled healthcare systems.	neural network.	detect and isolate suspect devices.	Real data.	99.4%.	- CPU usage and memory are not taken into account. -Not real-time.
[4] 2021	An Ensemble Learning (EL) algorithm based IDS that uses fog-cloud technology to mitigate cyberattacks in IoMT networks.	DT, NB, RF and XGBoost.	cyberattacks in IoMT environment.	ToN-IoT.	96.35%.	- CPU usage, battery and memory are not taken into account.
[17] 2021	An IDS based on ML methods which can distinguish between normal and attack traffic.	K-N N, R F, ANN, J48, SVM, and Decision Table.	MitM and DoS attacks.	Generated dataset.	97.67%.	- FPR, memory, and CPU utilization are not taken into account.

DL could be utilized within generative mode by unsupervised learning including AE (Autoencoder) and DBN (Deep Belief Network), discriminative mode employing supervised learning, for example CNN (Convolutional Neural Network) and RNN (Recurrent Neural Network), or a hybrid mode which combines both approaches, such as Generative Adversarial Network (GAN) and Ensemble of DL Networks (EDLNs) [13] [10] [30].

The recent advancements in DL technology have piqued the interest of many researchers in working on the IoT security issues utilizing DL models. Table III below provides a concise summary of recent research works conducted in order to propose IDS for IoMT using various DL-based techniques.

TABLE III. CURRENT DL-BASED IDS FOR IoMT.

Ref/year	Description	Algorithms	Attacks addressed	Dataset	Accuracy rate	Limitations
[55] 2016	iDetect, a multi-objective GA technique based IDS solution to provide optimum detection of attacks in WBANs.	Genetic Algorithm (GA).	Jamming, Selective forwarding, Deceptive jamming attacks.	Self-collected data.	93%.	- Training time and detection time not given. -Not lightweight.
[56] 2018	A secure data access and transmission in IoMT system using the DQN (Deep-Q-Network) approach.	Deep-Q Network (LDQN).	Malware detection.	Simulated data.	98.79%.	-High training overhead. -Memory usage and FPR are not taken into account.
[57] 2018	A MLP (Multilayer Perceptron) solution based on neural networks and embedded on a FPGA chip system to secure the treatment of diabetics in insulin pumps.	MLP-based neural network.	genuine vs fake glucose measurements.	UCI Diabetic dataset.	98.1%.	-High memory usage due to training.
[58] 2019	A DL approach based on LSTM (Long Short Term Memory) used for predicting various attack patterns in Deep Brain Stimulation.	Long Short Term Memory (LSTM).	impairment of motor functions, modification of impulsion control, alteration of affect or emotions.	Parkinson tremor dataset.	Not given.	-Not lightweight. -Accuracy not specified.
[59] 2020	A DNN to develop efficient and effective IDS in IoMT ecosystem to identify and prevent unforeseen cyberattacks.	DNN, PCAGWO (Principal Component Analysis and Grey Wolf Optimization).	Probe, U2R, R2L, and DoS.	Universal Kaggle dataset.	99.9%.	-High CPU and memory overhead. -Restricted to only IP-based devices.
[60] 2020	A DL-based model (DBN-IDS) to detect attacks in the IoMT.	DBN.	DoS/DDoS, Botnet, Brute Force, Infiltration, Web Attacks, and PortScan.	CICIDS 2017.	Varies from 96.37% to 98.37% for different attacks.	-Not real-time. -Not lightweight. - CPU and memory usage is not taken into account.
[26] 2021	A hybrid architecture SDN model (CNN-LSTM) driven by DL for malware detection in the IoMT.	CNN LSTM.	Malware attacks.	IoT Malware Dataset.	99.83%.	-High CPU and memory overhead. -Not real-time.

VII. DISCUSSION

The use of ML and DL algorithms in disease detection and management is inevitable in the healthcare industry. ML and DL applications can help medical practitioners by providing faster and more accurate solutions. They can improve the speed of decision making and reduce the false positive rate. ML and DL techniques can also be effectively applied to develop an intrusion detection system (IDS) for IoMT networks. This section discusses the major results and open research issues related to the ML/DL-based intrusion detection methods studied in the IoMT security context. Based on the analysis of the reviewed papers, we have noted that:

- A small part of studies employed lightweight methods [26] [46] [47], most of the proposed approaches require a high computational overhead for training their models, which is not affordable for IoMT devices as it will impact the energy consumption and IoMT devices are resource-constrained devices.

However, performances measures such as CPU, time, memory (space), and power overhead have been neglected.

- Most attack detection techniques have been conducted in an isolated off-line environment. Nevertheless, a simulation or hardware implementation should be considered to represent a real IoT environment.
- It can be seen that the most of the proposed IDS are addressing a predefined number of attacks or abnormal traffic behaviors, regardless the possible emergence of new types of attacks or the changing behavior that existing attacks may undergo over time. Moreover, they are assessed using either outdated datasets (e.g., NSL KDD or KDD CUP99) or datasets lacking modern IoMT system-based attacks.
- In terms of information sources and frequency of use, we note that the majority of proposals are focused on

real-time and off-line NIDS, with some expectations for HIDS as well as hybrid data auditing. Alternatively, we can observe that the strategy of placement is balanced, with an equal distribution of empirical and simulations validations. Generally, the propositions focus most on routing-based attacks, while a few addresses with preprocessing of raw data.

- In addition, few proposals cover appropriate preventive methods to prevent IoMT networks from external and internal attacks [26] [49] [59].

However, these intrusion detection systems are faced with various difficulties and challenges. One of these challenges is the issue of False Negatives (miscategorized non-abnormal packets) and False Positives (miscategorized attacks) in all attack detection approaches. An important factor to keep in mind when building a NIDS is that any intrusion that is misclassified as a non-intrusion is actually more hazardous than a false alert (non-intrusion categorized as an intrusion). Improving the accuracy of the IDS and reducing the FAR should not result in an increase in training and execution.

The ML techniques can build the model using limited data, can operate on conventional computers, needs less learning time, and runs on the conquer and divide or modularity principle. However, the DL approach needs a large volume of data to develop a model, requires more time than machine learning and runs on high performance computers [61].

The weak performance of conventional ML models (e.g., KNN, SVM, ANN) over massive datasets has led to a shift in research focus to apply advanced DL in building IDS [16]. As a result, many DL-based systems have been suggested for detecting attacks in the IoT. But, the appearance of numerous new sophisticated attacks has led to two additional challenges, namely, scalability and adaptability. Most deep learning-based solutions involve frequent model retraining and are difficult to scale. However, the use of DL for intrusion detection within an IoMT network is still in its early stages. More recently, researchers have begun looking for other ML methods such as reinforcement learning [62] to mitigate the two challenges mentioned above.

Computational complexity could be considered as one of the major obstacles in the context of deep learning and IoT security. The majority of DL algorithms are developed for training and use by resource-rich devices. Training a DL based algorithm with large amounts of data samples and training parameters needs devices that are resource rich in memory, computation and power. That is in stark contrast to the increasing interest in deploying resource-limited devices equipped with learning-based technologies.

Regarding the considerable progress in resource-limited IoMT devices and the significant increase of the data generated on the network's edge, it appears that edge computing will allow the distribution of various data processing resources over the cloud for analysis [63]. It is reasonable to assert that the incorporation of deep learning in IoT aids in the facilitation of security analytics by offering an improved processing system able of detecting attacks accurately and effectively.

Memory space is also another demand for many of the provided techniques, where it would gather the data analyzed and use it in an extensive analysis, however some IoMT

devices do not support advanced memory capacity but only offer very few kilobytes.

Real time ML methods are critical for IoMT systems, particularly in attacks and intrusions detection. Since recent IoMT devices comprise medical devices such as inhalers, pacemakers, among others, and if an attacker compromises these devices and the intrusion detection isn't performed in real time, it can turn out to be mortal, therefore a constant and rapid attack detection are required [14].

Time complexity is therefore critical for detecting threats within IoMT applications, since the existing detection techniques have been developed on the basis of batch processing applications instead of real time detection [64]. Furthermore, adoption of DL, particularly at the edge, necessitates the development of methods for distributing training and reducing training time. Additionally, DL is extremely capable of fixing time concerns in IoT by trying to implement GPU components to handle real time processing in an efficient way [65].

The quantity and quality of datasets used for building and training the models is critical to the success of IDS based on ML techniques, particularly DL algorithms. When operate in a live IoMT ecosystem, however, increased false negatives and false positives class labels would be returned [17]. When assessing novel IDS propositions, the lack of a proper dataset to implement ML/DL in IoMT security poses a significant challenge. Furthermore, the datasets used are sometimes not available to the public and are too small for appropriate use in DL. The majority of current research relies on prototype implementations, simulated data, and experimental testbeds. In this regard, there is a requirement for a security measure that addresses the more important set of attacks using recently updated datasets.

Another challenge to consider is the potential risk of adversary attacks on the ML/DL techniques themselves [12]. An adversary with sufficient knowledge of how ML/DL is working on data can tamper with the data on the training or the testing level to deceive the ML/DL methods. Therefore, the proposed approaches must be robust enough to withstand adversarial attacks.

While in the context of intrusion response and attack prevention, it is observed that additional work is required in this field to predict and assess the impacts of intrusions.

VIII. CONCLUSION

IoMT serves as health intelligence in smart healthcare projects for smart cities. It can benefit both doctors and patients by lowering costs, improving patient care, and reducing the workload of doctors and paramedics.

Despite its benefits, the IoMT is subject to different types of attacks, concerns, and challenges which primarily target the patients' privacy, integrity, confidentiality, and availability of healthcare services. As a result, ML/DL-based IDS must be implemented to provide a substantial, reliable, and stable network for the IoMT ecosystem.

In this work, a generic overview of intrusion detection methods, architectures, data sources, evaluation methods, and metrics is presented. Furthermore, we investigated different attack detection techniques for the IoMT network using ML and DL, highlighting the techniques used, accuracy, datasets, and some of the drawbacks. Moreover, by emphasizing open issues, current research challenges surrounding such

mechanisms, and future directions, this work attempts to motivate the researchers to deal with the challenges identified in this study in order to secure IoMT system with defensive techniques.

In the future, we intend to develop a new method for detecting different types of potential threats on IoMT system in the online phase as well as updating decision model with active learning.

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